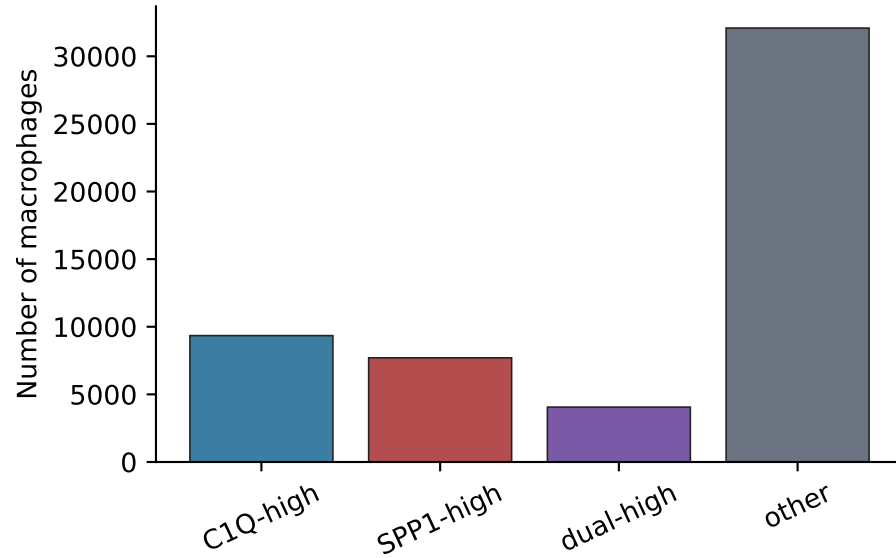
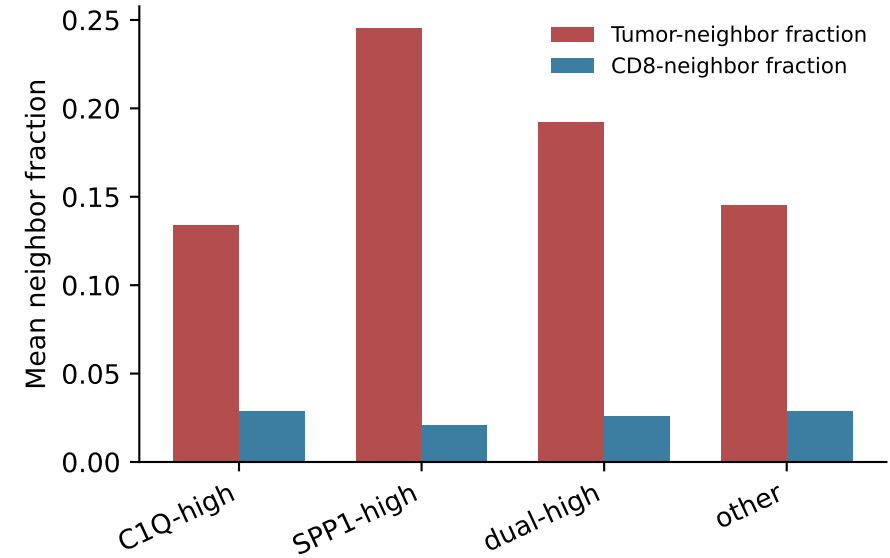


Selected CosMx macrophage-module evidence for C1Q/SPP1 spatial immune context

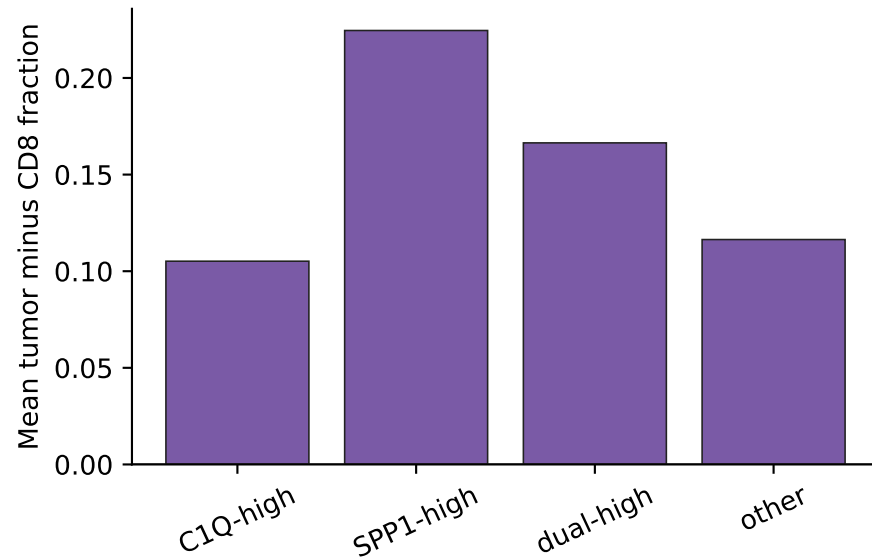
A. Official CosMx macrophage modules



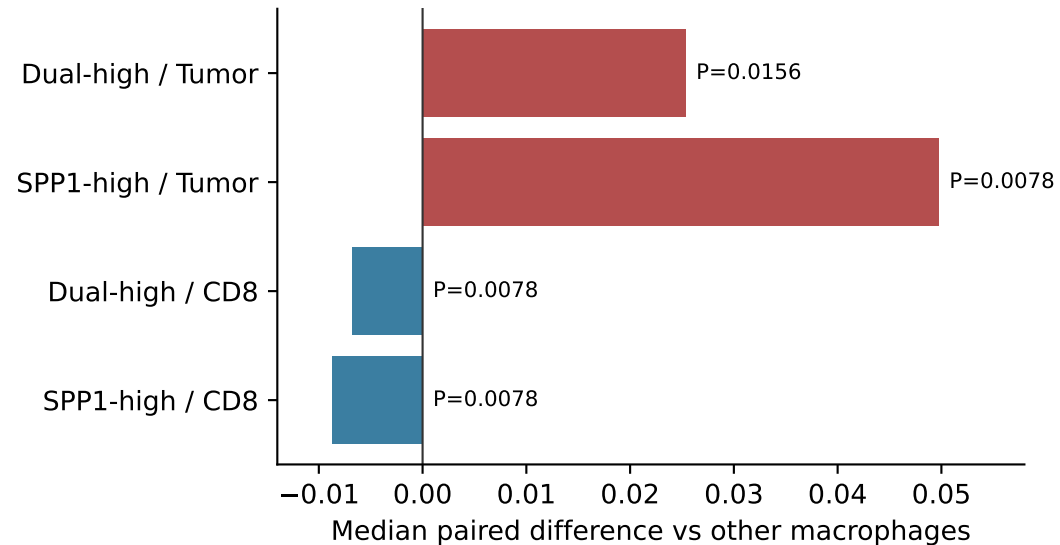
B. kNN100 tumor/CD8 neighborhood context



C. Tumor-minus-CD8 local context



D. Paired sample-level contrasts



Interpretation: tumor-neighborhood enriched and relatively CD8-lower context; not causal immune exclusion or functional suppression.